Scale Exercises - Section II, Level 16

Major Scale Four-Note Rolling Sequence: One-Octave Neck Ladder

Four forward; roll back two.

Learn the four-note overlap in the compact frame before extending it across two octaves.

1. Overlap four notes

Play 1-2-3-4, then start the next group on degree 2. Each group advances one degree while reusing three notes.

2. Lock the pick direction

Each group has four notes, so every group begins on the same stroke. Continue through crossings without resetting.

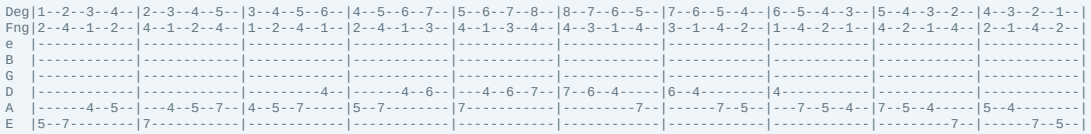
3. Shift only after the cell

Complete all 40 notes, return to the low root, release pressure, and move the complete frame exactly one fret.

Worked example - A major at low-E fret 5

Scale tones: A B C# D E F# G# A. The octave root is repeated at the turn.

UP: 1234 | 2345 | 3456 | 4567 | 5678 DOWN: 8765 | 7654 | 6543 | 5432 | 4321



KEY ROUTE A -> Bb -> B -> C -> Db -> D -> Eb -> E -> F -> Gb -> G -> Ab -> A, then reverse.

AUDIT Hear 10 equal four-note groups; keep overlap boundaries, crossings, and one-fret shifts invisible.

Returning to one octave is deliberate: isolate the four-note sequence first, then expand it in Level 17.

Session protocol

1. Master the A-major 40-note cell before adding a shift.
2. Use sixteenth notes: one four-note group per beat.
3. Start near 75% of your clean Level 15 BPM.
4. Repeat with a downstroke start and an upstroke start.
5. Use 2-3 neighboring rungs; pause one beat after each cell.
6. Remove the pause; connect the full route and reverse at fret 17.

Level 16 - Chromatic Major-Key Route and Sequence Map

Ascending route: complete one 40-note rolling cell, then move the frame +1 fret

Descending route: use the same cards in reverse order and move -1 fret

Rung	Key	Low-E root	Frame	Scale tones (degrees 1-7)
01	A major	fret 5	4-7	A B C# D E F# G#
02	Bb major	fret 6	5-8	Bb C D Eb F G A
03	B major	fret 7	6-9	B C# D# E F# G# A#
04	C major	fret 8	7-10	C D E F G A B
05	Db major	fret 9	8-11	Db Eb F Gb Ab Bb C
06	D major	fret 10	9-12	D E F# G A B C#
07	Eb major	fret 11	10-13	Eb F G Ab Bb C D
08	E major	fret 12	11-14	E F# G# A B C# D#
09	F major	fret 13	12-15	F G A Bb C D E
10	Gb major	fret 14	13-16	Gb Ab Bb Cb Db Eb F
11	G major	fret 15	14-17	G A B C D E F#
12	Ab major	fret 16	15-18	Ab Bb C Db Eb F G
13	A major (octave)	fret 17	16-19	A B C# D E F# G#

How to use the map

Up pass:
rungs 01-13.

Turn:
complete the high-A cell at fret 17, then reverse direction.

Down pass:
rungs 12-01. Do not replay the high-A card unless beginning a new trip.

Sixteenth pulse:
one four-note group per beat; never pause between groups.

Goal:
hear one connected scale line rather than 10 isolated fragments.

Sequence anatomy

```
UP:   1-2-3-4  |  2-3-4-5  |  3-4-5-6  |  4-5-6-7  |  5-6-7-8

DOWN: 8-7-6-5  |  7-6-5-4  |  6-5-4-3  |  5-4-3-2  |  4-3-2-1
```

Each new group begins one scale degree farther along; the boundary drops a diatonic third before moving forward again.

Mastery gate

Name every key, preserve the four-note pulse, land every one-fret shift silently, and complete the full route with both pick starts on two separate days.

Level 16 - Major Scale Four-Note Rolling Sequence

Rungs 01-03: play each card as one 10-group cell, then move +1 fret

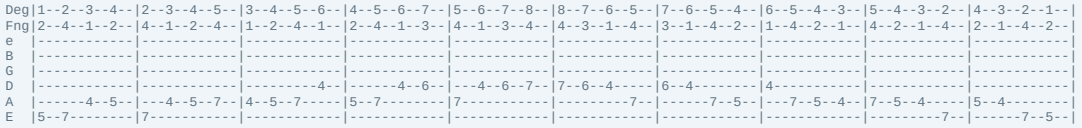
For the descending pass, read the cards in reverse order and move -1 fret

01 A MAJOR

ROOT E5 | FRAME 4-7

Scale tones: A B C# D E F# G# A

40-NOTE / 10-GROUP CELL



Return to the low root; release pressure; move the entire frame one fret.

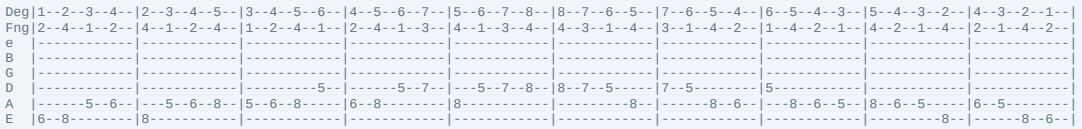
Four notes per beat - continuous alternate picking

02 BB MAJOR

ROOT E6 | FRAME 5-8

Scale tones: Bb C D Eb F G A Bb

40-NOTE / 10-GROUP CELL



Return to the low root; release pressure; move the entire frame one fret.

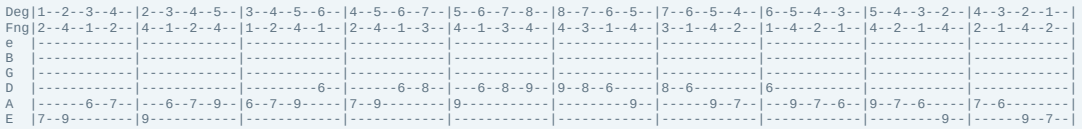
Four notes per beat - continuous alternate picking

03 B MAJOR

ROOT E7 | FRAME 6-9

Scale tones: B C# D# E F# G# A# B

40-NOTE / 10-GROUP CELL



Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

Level 16 - Major Scale Four-Note Rolling Sequence

Rungs 04-06: play each card as one 10-group cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

04 C MAJOR

ROOT E8 | FRAME 7-10

Scale tones: C D E F G A B C

40-NOTE / 10-GROUP CELL

```
Deg|1--2--3--4--|2--3--4--5--|3--4--5--6--|4--5--6--7--|5--6--7--8--|8--7--6--5--|7--6--5--4--|6--5--4--3--|5--4--3--2--|4--3--2--1--|
Fng|2--4--1--2--|4--1--2--4--|1--2--4--1--|2--4--1--3--|4--1--3--4--|4--3--1--4--|3--1--4--2--|1--4--2--1--|4--2--1--4--|2--1--4--2--|
e  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
B  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
G  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
D  |------------|------------|---------7--|------7--9--|---7--9--10-|10-9--7-----|9--7--------|7-----------|------------|------------|
A  |------7--8--|---7--8--10-|7--8--10----|8--10-------|10----------|---------10-|------10-8--|---10-8--7--|10-8--7-----|8--7--------|
E  |8--10-------|10----------|------------|------------|------------|------------|------------|------------|---------10-|------10-8--|
```

Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

05 DB MAJOR

ROOT E9 | FRAME 8-11

Scale tones: Db Eb F Gb Ab Bb C Db

40-NOTE / 10-GROUP CELL

```
Deg|1--2--3--4--|2--3--4--5--|3--4--5--6--|4--5--6--7--|5--6--7--8--|8--7--6--5--|7--6--5--4--|6--5--4--3--|5--4--3--2--|4--3--2--1--|
Fng|2--4--1--2--|4--1--2--4--|1--2--4--1--|2--4--1--3--|4--1--3--4--|4--3--1--4--|3--1--4--2--|1--4--2--1--|4--2--1--4--|2--1--4--2--|
e  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
B  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
G  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
D  |------------|------------|---------8--|------8--10-|---8--10-11-|11-10-8-----|10-8--------|8-----------|------------|------------|
A  |------8--9--|---8--9--11-|8--9--11----|9--11-------|11----------|---------11-|------11-9--|---11-9--8--|11-9--8-----|9--8--------|
E  |9--11-------|11----------|------------|------------|------------|------------|------------|------------|---------11-|------11-9--|
```

Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

06 D MAJOR

ROOT E10 | FRAME 9-12

Scale tones: D E F# G A B C# D

40-NOTE / 10-GROUP CELL

```
Deg|1--2--3--4--|2--3--4--5--|3--4--5--6--|4--5--6--7--|5--6--7--8--|8--7--6--5--|7--6--5--4--|6--5--4--3--|5--4--3--2--|4--3--2--1--|
Fng|2--4--1--2--|4--1--2--4--|1--2--4--1--|2--4--1--3--|4--1--3--4--|4--3--1--4--|3--1--4--2--|1--4--2--1--|4--2--1--4--|2--1--4--2--|
e  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
B  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
G  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
D  |------------|------------|---------9--|------9--11-|---9--11-12-|12-11-9-----|11-9--------|9-----------|------------|------------|
A  |------9--10-|---9--10-12-|9--10-12----|10-12-------|12----------|---------12-|------12-10-|---12-10-9--|12-10-9-----|10-9--------|
E  |10-12-------|12----------|------------|------------|------------|------------|------------|------------|---------12-|------12-10-|
```

Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

Sixteenth notes: one four-note group per beat. Do not restart alternate picking at a group boundary.

Level 16 - Major Scale Four-Note Rolling Sequence

Rungs 07-09: play each card as one 10-group cell, then move +1 fret

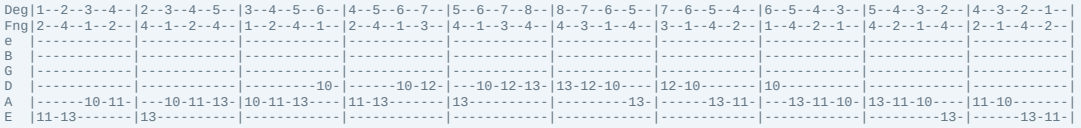
For the descending pass, read the cards in reverse order and move -1 fret

07 EB MAJOR

ROOT E11 | FRAME 10-13

Scale tones: Eb F G Ab Bb C D Eb

40-NOTE / 10-GROUP CELL



Return to the low root; release pressure; move the entire frame one fret.

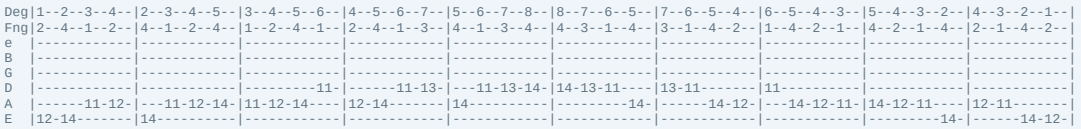
Four notes per beat - continuous alternate picking

08 E MAJOR

ROOT E12 | FRAME 11-14

Scale tones: E F# G# A B C# D# E

40-NOTE / 10-GROUP CELL



Return to the low root; release pressure; move the entire frame one fret.

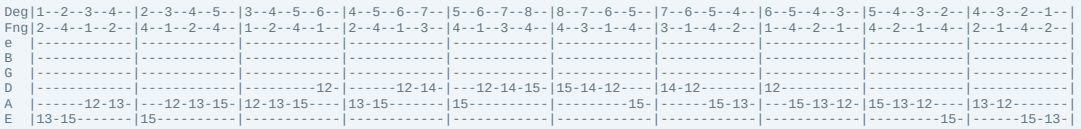
Four notes per beat - continuous alternate picking

09 F MAJOR

ROOT E13 | FRAME 12-15

Scale tones: F G A Bb C D E F

40-NOTE / 10-GROUP CELL



Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

Sixteenth notes: one four-note group per beat. Do not restart alternate picking at a group boundary.

Level 16 - Major Scale Four-Note Rolling Sequence

Rungs 10-12: play each card as one 10-group cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

10 GB MAJOR

ROOT E14 | FRAME 13-16

Scale tones: Gb Ab Bb Cb Db Eb F Gb

40-NOTE / 10-GROUP CELL

```
Deg|1--2--3--4--|2--3--4--5--|3--4--5--6--|4--5--6--7--|5--6--7--8--|8--7--6--5--|7--6--5--4--|6--5--4--3--|5--4--3--2--|4--3--2--1--|
Fng|2--4--1--2--|4--1--2--4--|1--2--4--1--|2--4--1--3--|4--1--3--4--|4--3--1--4--|3--1--4--2--|1--4--2--1--|4--2--1--4--|2--1--4--2--|
e  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
B  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
G  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
D  |------------|------------|---------13-|------13-15-|---13-15-16-|16-15-13----|15-13-------|13----------|------------|------------|
A  |------13-14-|---13-14-16-|13-14-16----|14-16-------|16----------|---------16-|------16-14-|---16-14-13-|16-14-13----|14-13-------|
E  |14-16-------|16----------|------------|------------|------------|------------|------------|---------16-|------16-14-|
```

Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

11 G MAJOR

ROOT E15 | FRAME 14-17

Scale tones: G A B C D E F# G

40-NOTE / 10-GROUP CELL

```
Deg|1--2--3--4--|2--3--4--5--|3--4--5--6--|4--5--6--7--|5--6--7--8--|8--7--6--5--|7--6--5--4--|6--5--4--3--|5--4--3--2--|4--3--2--1--|
Fng|2--4--1--2--|4--1--2--4--|1--2--4--1--|2--4--1--3--|4--1--3--4--|4--3--1--4--|3--1--4--2--|1--4--2--1--|4--2--1--4--|2--1--4--2--|
e  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
B  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
G  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
D  |------------|------------|---------14-|------14-16-|---14-16-17-|17-16-14----|16-14-------|14----------|------------|------------|
A  |------14-15-|---14-15-17-|14-15-17----|15-17-------|17----------|---------17-|------17-15-|---17-15-14-|17-15-14----|15-14-------|
E  |15-17-------|17----------|------------|------------|------------|------------|------------|---------17-|------17-15-|
```

Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

12 AB MAJOR

ROOT E16 | FRAME 15-18

Scale tones: Ab Bb C Db Eb F G Ab

40-NOTE / 10-GROUP CELL

```
Deg|1--2--3--4--|2--3--4--5--|3--4--5--6--|4--5--6--7--|5--6--7--8--|8--7--6--5--|7--6--5--4--|6--5--4--3--|5--4--3--2--|4--3--2--1--|
Fng|2--4--1--2--|4--1--2--4--|1--2--4--1--|2--4--1--3--|4--1--3--4--|4--3--1--4--|3--1--4--2--|1--4--2--1--|4--2--1--4--|2--1--4--2--|
e  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
B  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
G  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
D  |------------|------------|---------15-|------15-17-|---15-17-18-|18-17-15----|17-15-------|15----------|------------|------------|
A  |------15-16-|---15-16-18-|15-16-18----|16-18-------|18----------|---------18-|------18-16-|---18-16-15-|18-16-15----|16-15-------|
E  |16-18-------|18----------|------------|------------|------------|------------|------------|---------18-|------18-16-|
```

Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

Sixteenth notes: one four-note group per beat. Do not restart alternate picking at a group boundary.

Level 16 - Final Rung and Mastery Work

Rung 13: complete the octave A-major cell, then reverse the neck route

13 A MAJOR (OCTAVE)

ROOT E17 | FRAME 16-19

Scale tones: A B C# D E F# G# A

40-NOTE / 10-GROUP CELL

```
Deg|1--2--3--4--|2--3--4--5--|3--4--5--6--|4--5--6--7--|5--6--7--8--|8--7--6--5--|7--6--5--4--|6--5--4--3--|5--4--3--2--|4--3--2--1--|
Fng|2--4--1--2--|4--1--2--4--|1--2--4--1--|2--4--1--3--|4--1--3--4--|4--3--1--4--|3--1--4--2--|1--4--2--1--|4--2--1--4--|2--1--4--2--|
e  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
B  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
G  |------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
D  |------------|------------|---------16-|------16-18-|---16-18-19-|19-18-16----|18-16-------|16----------|------------|------------|
A  |------16-17-|---16-17-19-|16-17-19----|17-19-------|19----------|---------19-|------19-17-|---19-17-16-|19-17-16----|17-16-------|
E  |17-19-------|19----------|------------|------------|------------|------------|------------|---------19-|------19-17-|
```

Return to the low root; release pressure; move the entire frame one fret.

Four notes per beat - continuous alternate picking

Full round-trip challenge

Start at A major on low-E fret 5. Play rungs 01-13 without restarting the pick cycle. Reverse after the fret-17 A-major cell and return through rungs 12-01. Say each key name aloud. Repeat with the opposite starting stroke.

CLEAN-ROUND STANDARD

No missing group, wrong scale tone, pick reset, open string, shift scrape, crossing accent, or unequal repeated octave roots. The 25-cell trip contains exactly 1,000 picked notes.

After the basic ladder is clean

1. Accent note 1 of every group, then remove all accents.
2. Move the accent to notes 2, 3, and 4 on later passes.
3. Accent only degrees 1 and 8 wherever they appear.
4. Begin at fret 17 and descend first.
5. Enter at a random root fret, name the key, and continue.

These are variations, not separate full levels.

Practice log

Date	BPM	Start stroke	Clean route	Notes